a **The 030 grid: every (size, subset) at seed 1, replicates where they are cheap, the 025 pair at the current size**

columns: nested S3Q subsets rows: size and N	1/64: 6 pairs 21k records, 33k gi rows	1/16: 21 pairs 69k records, 132k gi rows	1/4: 83 pairs 244k records, 509k gi rows	full: 331 pairs 1.05M records, 2.34M gi rows
45 x 2: N 79k	●●●3 seeds	●●●3 seeds	● ○ 1 seed + lr 1e-4, 1e-3	● 1 seed
63 x 3: N 203k	● 1 seed	● 1 seed	● 1 seed	● 1 seed
90 x 4: N 509k	● 1 seed	● 1 seed	● 1 seed	● 1 seed
126 x 6: N 1.38M	● 1 seed	● 1 seed	● 1 seed	● 1 seed
180 x 8: N 3.59M	●●●3 seeds	●●●3 seeds	● 1 seed	● 1 seed, 10 epochs: headline
025 pair, 180 x 8 merged label	■■■3 seeds	■■■3 seeds	■ 1 seed	■ 1 seed
epochs (one 48 h job)	240	120	40	10
records seen	5.12M	8.26M	9.78M	10.46M
hours at 68 records/s (assumed)	21	34	40	43

matched read at B = the smallest total any point reaches, 5.12M records (the 1/64 total); every point is read there

calibration first: 400 batches at 180 x 8 and at 45 x 2 on the full pool, 3 h each. Equal paces mean the loader binds and no size finishes sooner than 180 x 8.

● 030 per-entry arm, dataset token (seed 1)

■ 025 merged-label twin, same subsets

○ learning-rate check at the smallest size

every job: batch 256, lr 2.5e-4 constant, one A40, full-pool normalization constants